National Real-Time Payments System : Architectural Discussion Questions and Response

Q1. What are the various systems and subsystems required to deliver this functionality?

- User onboarding system with KYC
- Bank account linking subsystem
- Real-time P2P and P2M payment processing engine
- QR-based payment subsystem
- Transaction history and receipt management
- Transaction status service (success, pending, failed)
- Integration layer with major banks
- Inter-bank settlement and reconciliation system
- Scheduled settlement subsystem
- Dispute and chargeback handling system
- Merchant registration and verification system
- Merchant dashboards and daily settlement reports
- Admin dashboards
- Regulatory reporting subsystem
- Audit logs
- Fraud detection and risk scoring engine

Q2. What kind of information is exchanged between systems?

- KYC information during onboarding
- Bank account details for linking
- Real-time payment request and response data
- QR code payment information
- Transaction history and receipts
- Transaction status updates
- Inter-bank settlement and reconciliation data
- Scheduled settlement files
- Dispute and chargeback information
- Merchant registration, verification, and settlement data
- Audit log entries
- Fraud detection and risk scoring data

Q3. What are the major potential failure points in this system?

- Technical failures: latency issues, availability issues, engine overload
- Integration failures: bank API downtime, merchant POS failures
- External dependency failures: partial bank outages, national network issues

Q4. What mitigation strategies would you design to prevent or reduce failures?

- Redundancy to avoid single points of failure
- Circuit breakers for failing systems
- Horizontal scalability to handle 10x load
- Automatic failover mechanisms
- Graceful degradation
- Asynchronous processing

- Real-time monitoring and alerting
- Zero data loss replication techniques

Q5. What contingency plans are required when failures do occur?

- Manual reconciliation
- Deferred settlements
- User communication strategies
- Failover routing to secondary infrastructure

Q6. How should the system behave under normal daily load vs peak national events?

- Normal load: standard fraud checks and normal operations
- Peak load: horizontal scaling, maintain <2 sec latency, operate during partial bank outages

Q7. What architectural trade-offs would you make?

- Consistency vs availability: ensure availability for payments
- Cost vs reliability: maintain reliability with scaling and redundancy
- Real-time vs eventual: settlements can be scheduled
- Security depth vs user experience: comply with PCI DSS & ISO 27001

Q8. Global implementations, USPs, and ROI considerations

- Currency conversion support
- Cross-border settlement capabilities
- Integration with foreign banking systems
- Compliance with international financial regulations